

Data Science 6

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Pivots and Joins

0.0.1 Review

Recall the grouping method, can group by single or multiple column or two columns (**visulaztion tool**). Rows are sorted in ascending order when using group function (for strings this means alphabetically.) Recall tricks like dropping columns and renaming columns from previous hws.

0.0.2 Pivoting

Pivot is a function that involvs two categorical vars. The **first** argument is the name of the column and the values of that column are taken and turned into columns (no duplicates). The **second** arguments values turns into rows and the table itself is the counts of the intersections. May seem a little wordy but to say it simply its better look at the following image.



Flavor	Color	Price
strawberry	pink	3.55
chocolate	light brown	4.75
chocolate	dark brown	5.25
strawberry	pink	5.25
chocolate	dark brown	5.25
bubblegum	pink	4.75

Color	bubblegum	chocolate	strawberry
dark brown	0	2	0
light brown	0	1	0
pink	1	0	2

Figure 1. Example

Definition 0.0.1: `table_name.pivot(col_values, row_values, values, collect)`

Builds a contingency table from two categorical variables: one column's values become the new columns, another's become the rows, and each cell aggregates the intersecting entries.

col_values column name whose values become the columns

row_values column name whose values become the rows

values column whose entries fill the body of the table

collect function describing how to combine those entries

0.0.3 Joins

We can combine info from different tables via joining. We essintaly join using values of a ceartin column and we specify what we want for rows and what we want for columns. When there exist a row within the table we are calling that has no value in the other table it is dropped. Last argument is optional only if columns that are joining on has a diffrent name. GIVE ALL COMBINATIONS POSSIBLE.

Note:-

The first argument is the column from tblA we are joining on and the last column argument is the column from tblB we are joining on.

Definition 0.0.2: `tblA.join(colA, tblB, colB)`

Combines information from two tables by matching on a shared column. Rows in `tblA` with no matching value in `tblB` are dropped. The final argument is only needed when the two join columns have different names.

`colA` column in `tblA` to join on

`tblB` the table whose rows are joined in

`colB` column in `tblB` to join on